

Department of Physics, Princeton University

Graduate Preliminary Examination
Part I

Thursday, January 9, 2025

9:00 am - 12:00 noon

Answer TWO out of the THREE questions in Section A (Mechanics) and TWO out of the THREE questions in Section B (Electricity and Magnetism).

Work each problem in a separate examination booklet.

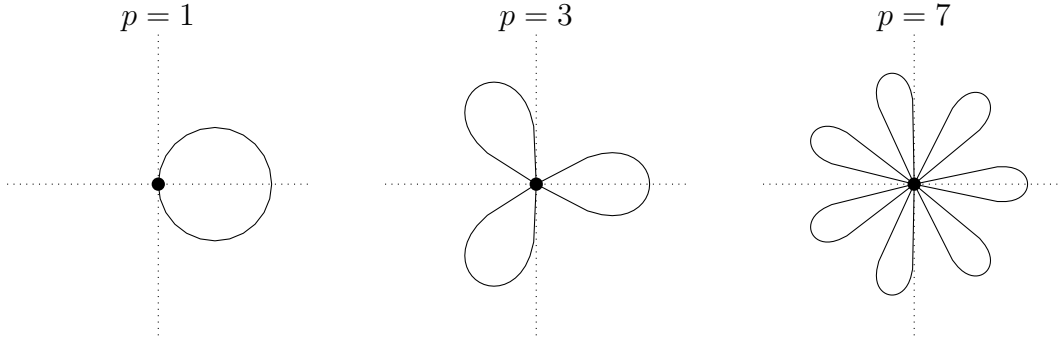
Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Mechanics

1. Rose orbits

A mass m is moving under the influence of a central force directed towards the origin at $r = 0$. Its angular momentum about the origin is L , and use θ to denote the corresponding angular coordinate.

a) Write the radial equation of motion in terms of the force $F(r)$. Then express $F(r)$ in terms of $L, m, u \equiv 1/r$ and $\frac{d^2u}{d\theta^2}$.



b) We want the orbit the mass follows to take the shape of a rose with p petals, described by the polar equation

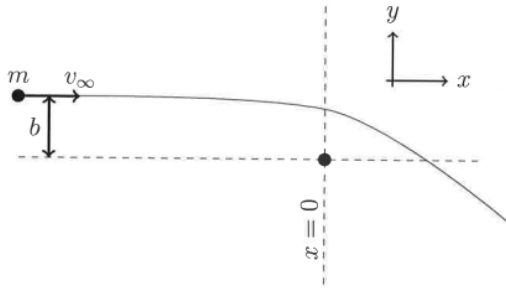
$$\left(\frac{r(\theta)}{R}\right)^p = \cos(p\theta), \quad (1)$$

where R is a fixed length scale and $p > 0$ is an odd integer. Determine the potential energy function $V(r)$ associated with the force $F(r)$ needed to accomplish this, and write your answer in terms of r, L, R, m , and p .

c) Let the area of a single petal be $a_p R^2$, where a_p is a numerical constant (e.g. $a_1 = \frac{\pi}{4}$). Determine the period T of the orbit in terms of L, R, m, p , and a_p .

2. Critical impact parameter.

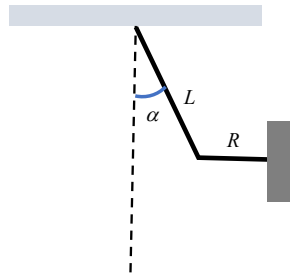
An object with mass m_1 and radius R_1 interacts with an object of mass m_2 and radius R_2 via an attractive potential, $V(r) = -\frac{k}{r}$. The motion is non-relativistic in three dimensions. The object of mass m_1 has a speed v_∞ relative to m_2 while the separation is very large, and approaches with an impact parameter b , as shown in the diagram.



- a) Compute the closest approach of the two objects, d , as a function of the given parameters.
- b) Find the critical value of the impact parameter b_c , such that for $b < b_c$ the objects will collide. Find the limiting values of b_c for $v_\infty \rightarrow 0$ and $v_\infty \rightarrow \infty$.

3. Gyroscope on a string

A gyroscope wheel is at one end of a massless axle of length R . The other end is attached to a massless, inextensible string of length L that is fixed to the ceiling. The wheel is set into motion such that it precesses at a uniform rate Ω in the horizontal plane, around the vertical. The wheel has mass M , moment of inertia I_0 about its axle that runs through its center of mass, and spin angular velocity ω_s about its axle (with ω_s sufficiently large that this steady horizontal precession occurs).



- What is the precession rate Ω in the horizontal plane?
- Find the angle α that the string makes with the vertical, assuming the small-angle approximation.

Section B. Electricity and Magnetism

1. Moving Magnetic Monopole and an Electric Charge

A magnetic monopole with magnetic charge q_m is moving with a constant velocity $\mathbf{v}$ along the x -axis. An electric charge q_e is held stationary at a distance d from the path of the monopole, along the y -axis. Assume $v \ll c$ and ignore corrections of order $(v/c)^2$ or higher.

- (a) Find the magnetic field due to the monopole at the position of the stationary electric charge as a function of time t . Choose $t = 0$ to be the time that the monopole is closest to the electric charge.
- (b) Find the electric field due to the monopole at the position of the stationary electric charge as a function of time t .
- (c) Find the magnitude and direction of the force on the stationary electric charge as a function of time.
- (d) At what time $t = t_{max}$ is the force greatest and what is its magnitude?

2. Charged Wires

Two thin infinite wires with charge per unit length λ and $-\lambda$ are placed parallel to each other.

(a) Show that equipotential surfaces are cylinders and a plane (Note: $\int \frac{dx}{\sqrt{1+x^2}} = \log(x + \sqrt{1+x^2}) + \text{constant}$).

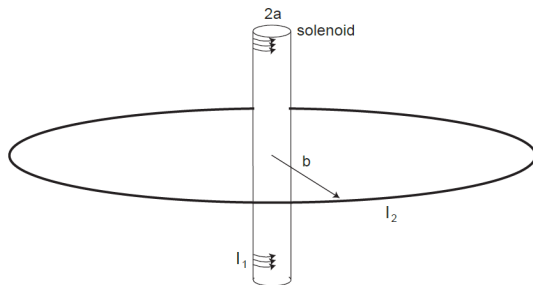
(b) The two wires are removed and replaced with an infinite conducting plate and an infinite conducting cylinder of radius r whose axis is at a distance d parallel to the plate ($d > r$). The plate and cylinder are maintained at potentials 0 and V_0 respectively ($V_0 > 0$). Find the charge per unit length on the cylinder as a function of d, r and V_0 .

3. A Solenoid

Consider a very long solenoid of radius a that carries a current $I_1(t)$, which decays linearly in time. The solenoid is concentric with a large circular loop (loop 2) of radius b ($b \gg a$). The loop is comprised of a single wire with total resistance R . Due to the decay of $I_1(t)$, the field inside the solenoid decreases as:

$$B(t) = B_0 - \alpha t.$$

As a result, a current I_2 is induced in loop 2.



- Find the induced current I_2 (in terms of the parameters mentioned).
- Compare the power dissipated in loop 2 with the power radiated from the solenoid, as computed through the Poynting vector:

$$\mathbf{S} = \frac{\mathbf{E} \times \mathbf{B}}{\mu_0},$$

integrated over the cylindrical surface enclosing the solenoid.

Hint: On that surface, where do $\mathbf{E}$ and $\mathbf{B}$ come from?

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Graduate Preliminary Examination
Part II

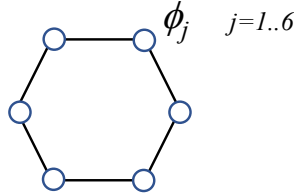
Friday, January 10, 2025
9:00 am - Noon

Answer TWO out of the THREE questions in Section A (Quantum Mechanics) and TWO out of the THREE questions in Section B (Thermodynamics and Statistical Mechanics).

Work each problem in a separate booklet. Be sure to label each booklet with your name, the section name, and the problem number.

Section A. Quantum Mechanics

1. A Benzene Molecule



The Benzene molecule can be modeled as a collection of six identical atomic orbitals $|\phi_j\rangle$, ($j = 1, \dots, 6$), each centered at the corners of a hexagon, as shown in the figure. The orbitals form an orthonormal basis set, in terms of which any single particle state can be represented, i.e. $|\psi\rangle = \sum_{j=1}^6 c_j |\phi_j\rangle$. The Hamiltonian for the single particle states can be written as a matrix in the above basis set:

$$H \longleftrightarrow \begin{bmatrix} E_0 & -A & 0 & 0 & 0 & -A \\ -A & E_0 & -A & 0 & 0 & 0 \\ 0 & -A & E_0 & -A & 0 & 0 \\ 0 & 0 & -A & E_0 & -A & 0 \\ 0 & 0 & 0 & -A & E_0 & -A \\ -A & 0 & 0 & 0 & -A & E_0 \end{bmatrix} \quad (1)$$

where E_0 is a constant energy, and $A > 0$ is a real hopping parameter.

a) The molecule has a 6-fold rotational symmetry operator R , which commutes with the Hamiltonian H . Thus, an energy eigenstate $|\Psi_n\rangle = \sum_{j=1}^6 c_{n,j} |\phi_j\rangle$ can be chosen as simultaneous eigenstate of the rotation symmetry R , where $c_{n,j}$ are the coefficients, and n is a proper label of the eigenstates. Use the rotational symmetry properties of the molecule to determine the coefficients $c_{n,j}$ of all the eigenstates (namely, find the eigenstates of R). Make sure the coefficients $c_{n,j}$ are such that the eigenstates are normalized, $\langle \Psi_n | \Psi_n \rangle = 1$.

b) Find the energy eigenvalues ϵ_n in terms of the constant E_0 and hopping parameter A . Draw a diagram for the energy levels and mark any degeneracies.

c) Assume the electrons in this structure are not interacting. Find the energies E_{GS} and E_{1st} of the ground state and the first excited state for 6 electrons occupying the energy levels of part 2, and draw a diagram to represent the states. (Recall that each electron orbital can be occupied by 2 spins.)

d) *next page ...*

d) If we had only one atomic orbital (instead of six), we know that the energy of a single electron occupying this orbital is E_0 . A single electron occupying the lowest state of the above Hamiltonian with 6 orbitals has a lower energy. Why? Can you give a physical reason?

2. Quantum Quenches

A particle with mass m is initially in a harmonic oscillator with angular frequency ω . The system is describe by the Hamiltonian

$$\hat{H}_0 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2, \quad (2)$$

where $\hat{x}$ and $\hat{p}$ are position and momentum operators, respectively. You may find the following identities for annihilation and creation operators in a quantum harmonic oscillator useful:

$$\begin{aligned} \hat{a} &= \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + i\frac{\hat{p}}{m\omega} \right) \\ \hat{a}^\dagger &= \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - i\frac{\hat{p}}{m\omega} \right) \end{aligned}$$

- a) What is the spectrum of the $\hat{H}_0$? That is, provide an expression for the energy E_n of the n^{th} eigenstate $|n\rangle$, where $n = 0$ denotes the ground state. (No derivation needed.)
- b) The particle is initially prepared in eigenstate $|n\rangle$. Compute $\langle n|\hat{x}^2|n\rangle$.
- c) With the particle initially in $|n\rangle$, the frequency is suddenly changed from ω to $\alpha\omega$, with $\alpha > 0$. The new Hamiltonian is given by

$$\hat{H}_1 = \frac{\hat{p}^2}{2m} + \frac{1}{2}m(\alpha\omega)^2\hat{x}^2. \quad (3)$$

What is the total energy E_{tot} of the particle after the quench from $\hat{H}_0$ to $\hat{H}_1$?

- d) Suppose that the particle is prepared initially in $|0\rangle$, the ground state of $\hat{H}_0$, before the quench to $\hat{H}_1$. After the quench, what is the probability P that it is in $|0'\rangle$, the ground state of $\hat{H}_1$?

3. Particle with a delta function potential

A particle of mass m moves on a line under an attractive delta function potential, with the Hamiltonian

$$H_\lambda = \frac{1}{2m}p^2 - \lambda\delta(x) \quad (4)$$

(a) Present a variational argument proving that for any $\lambda > 0$, the operator H_λ has at least one eigenstate of negative energy.

Hint: It suffices to consider simple test functions, such as normalized versions of the “tent function”

$$\psi_R(x) = \begin{cases} 1 - \frac{|x|}{R} & \text{if } |x| \leq R, \\ 0 & \text{if } |x| > R. \end{cases}$$

(b) What would you expect the particle’s ground state wave function to be under reflections ($x \rightarrow -x$): odd, even, or neither? (State the relevant general principle, in case one comes to mind.)

(c) Determine the ground state energy E_0 of the Hamiltonian H_λ through the computation of the corresponding normalized ground state eigenfunction $\psi_0(x)$.

Hint: The delta potential in the Schrödinger equation for the eigenfunction $\psi(x)$ forces discontinuity in the derivative $\frac{\partial}{\partial x}\psi(x)$ at $x = 0$.

(d) We denote the above ground state as $|\psi_0\rangle$. Assume at time $t < 0$, the system has Hamiltonian H_λ and the particle is in its ground state $|\psi_0\rangle$. At time $t \geq 0$, the Hamiltonian is suddenly changed to $H_{2\lambda}$ given by

$$H_{2\lambda} = \frac{1}{2m}p^2 - 2\lambda\delta(x) , \quad (5)$$

namely, the potential λ doubles into 2λ . Denote the new ground state of $H_{2\lambda}$ as $|\tilde{\psi}_0\rangle$. As time $t \rightarrow +\infty$, find out the probability P_0 that the particle stays in the new ground state $|\tilde{\psi}_0\rangle$.

Section B. Statistical Mechanics and Thermodynamics

1. A Debye Solid

An article in the Jan 7, 2020 NYT science section was entitled “The Superpowers of Super-Thin Materials. In materials science, 2-D is the new 3-D.” Although it focused on molybdenum disulfide, it also mentioned graphene, a two-dimensional single-atomic-layer-thick sheet of carbon atoms. In the following, model the graphene as a Debye solid with an isotropic speed of sound, c_s , with two polarizations.

- a) What is the density of states, g_ω , in two dimensions?
- b) For a 2D system with N particles, what is the Debye frequency, ω_D ?
- c) What is the heat capacity at constant area, C_A , associated with the lattice vibrations in the low temperature limit, $\hbar\omega_D \gg kT$? You may leave your answer in terms of a dimensionless integral.
- d) In graphene one might have “spin waves” as well as sound waves. Instead of $E = \hbar\omega(k) = \hbar c_s k$, where k is the magnitude of the wave vector, one has $\omega(k) = Jk^2$. In this case, how does C_A scale with temperature at low temperatures as compared to what one finds in part (c)?

2. Multi-level quantum systems at thermal equilibrium

Consider a system consisting of N quantum mechanical degrees of freedom (“molecules”). Each molecule can take $g + 1$ quantum mechanical states, where $g > 1$ is an integer. Among these levels, g of them are degenerate; we set the energy of these states $= 0$. The energy of the remaining level is ε . Molecules do not interact with each other and are all independent. We assume that the system is at thermal equilibrium at temperature T , i.e., at the corresponding inverse temperature $\beta = 1/k_B T$ where k_B is the Boltzmann constant.

- a) Calculate the thermal average of the energy.
- b) Calculate the entropy. Consider the three cases $\varepsilon > 0$, $\varepsilon = 0$, $\varepsilon < 0$ separately, and calculate the entropy at $T = 0$. Explain why the results are different.
- c) Let us now “distort” the molecules by some external force. This results in changing ε adiabatically, $\varepsilon \rightarrow \varepsilon'$. Calculate the change in temperature in this adiabatic process.
- d) (We now go back to the original molecules with the excited state energy level ε .) Let N_ε be the number of molecules having energy ε . Calculate the probability distribution of N_ε at thermal equilibrium, $P(N_\varepsilon)$. *Hint:* it may be useful (but optional) to recall the binomial expansion, $(x + y)^N = \sum_{m=0}^N {}_N C_m x^{N-m} y^m$ where ${}_N C_m = \binom{N}{m} = \frac{N!}{m!(N-m)!}$ is the binomial coefficient.
- e) Instead of using the canonical ensemble, let us use the microcanonical ensemble. Calculate the number of states $W(E)$ with energy $E = N_\varepsilon \varepsilon$. Assuming both N and N_ε are large, use the Stirling formula $\ln(n!) \simeq n \ln n - n$ for large n and express the Boltzmann entropy $S = k_B \ln W(E)$ in terms of N and E . Also, calculate temperature as a function of E from the microcanonical entropy $S(E)$. *Hint:* you may want to compare with the thermal average of the energy calculated in Part (a) using the canonical ensemble.

3. Cooling by adiabatic demagnetization

In this problem, you will study cooling by adiabatic demagnetization in spin systems, used e.g. in the context of nuclear magnetic resonance. Consider an ensemble of N spins $1/2$ in a solid, subject to an external magnetic field B along the z direction, such that the Hamiltonian of the system is given by:

$$H = - \sum_{i,j}^N \sum_{\alpha\beta=x,y,z} J_{i,j}^{\alpha\beta} \sigma_i^\alpha \sigma_j^\beta - \mu B \sum_{i=1}^N \sigma_i^z. \quad (6)$$

The first term describes dipolar interactions between spins, and let us assume that the typical value of the interaction for closest pairs of spins is given by

$$J_{ij} \sim \mu h. \quad (7)$$

a) First, let us assume that the system's temperature $T_i \gg h$ and that the external field is stronger than the interaction field, $B = B_i \gg h$. Compute the free energy, entropy, and magnetic susceptibility of the spin system.

b) Now, consider the following cooling protocol: (i) The system is initially subject to a magnetic field B_i and is in contact with a thermal bath with temperature T_i . (ii) We increase the field to a value $B_f > B_i$, while the system remains in contact with the bath. (iii) We disconnect the system from the bath, and reduce the field back to B_i . The system is then allowed to reach thermal equilibrium while remaining in isolation (which can be established e.g. due to interactions between spins).

What is the new temperature of the spin system? Sketch the entropy-temperature diagram of the system for different values of B and illustrate the trajectory of the system during cooling. Discuss various limiting cases and identify parameter regimes where cooling is efficient.

c) Next, let us take the interactions between spins into account. At a qualitative level, how would they modify the entropy-temperature relations you sketched above? At which values of B, T would the interactions have a significant effect?

d) To model the effect of interactions on cooling, let us first assume that $B_i = 0$. For simplicity, let us also approximate the effect of interactions as simply changing the magnetic field experienced by each spin, according to a relation

$$B_{\text{eff}} = \sqrt{B^2 + h^2}. \quad (8)$$

Assuming that the maximum field $B_f = C \times T_i / \mu$, with $C > 1$ a constant, what would the lowest attainable temperature of the spin system be?